

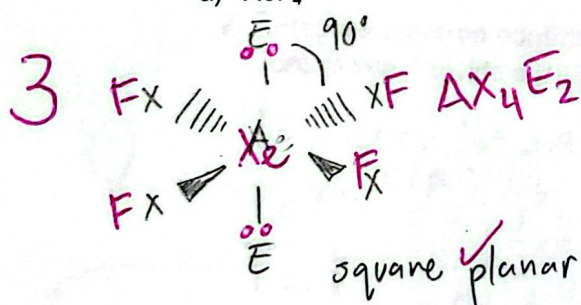
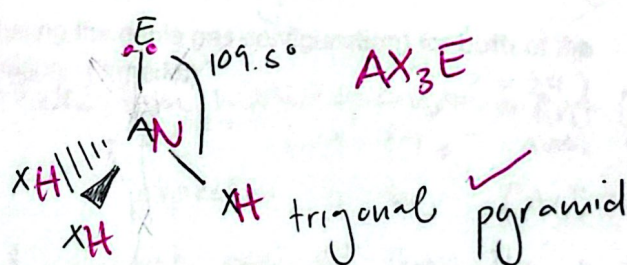
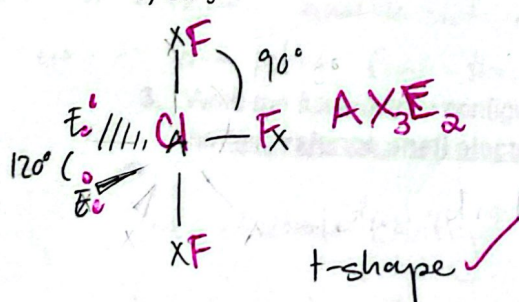
Name: _____

Date: _____

VSEPR Quiz

1. Identify the molecular geometry of each of the following (5 marks – 0.5 each):

Chemical Formula	Electronic Geometry name	Bond Angles
PF ₅	trigonal bipyramid ✓	120°; 90° ✓
SCl ₆	octahedral ✓	90° ✓
PCl ₄ ⁺	tetrahedral ✓	109.5° ✓
CO ₂	linear ✓	180° ✓
CO ₃ ²⁻	trigonal planar ✓	120° ✓

2. Draw the VSEPR shapes for the following molecules. Include the VSEPR molecular geometry name and the AXE formula. (2 marks each)a) XeF₄b) NH₃c) ClF₃

bonds repulse each other away)

3. What is the difference between Lewis dot structures and VSEPR models? (1 mark)

The difference between Lewis dot and VSEPR models is that Lewis dot diagrams are 2D, simpler diagrams of molecules. VSEPR models represent molecules on a 3D plane and mathematically calculate the bond angles between bonds and each other or lone pairs. They account for valence shell electron pair repulsion (e⁻ and